Ordinary Differential Equations

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1 Introduction

An ordinary differential equation (ODE) is an equation involving an unknown function and its ordinary derivatives with respect to **a single independent variable**. The goal of solving an ODE is to find a function that satisfies the given equation.

For example, consider

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \cos(t),$$

we know

$$x(t) = \sin(t) + C,$$

where C is an arbitrary constant. This example illustrates an important point that knowing the derivative of a function does not necessarily determine the function **uniquely**. Now, suppose we given the initial condition

$$x(0) = 2.$$

Substituting $t = 0$ into the general solution gives

$$x(0) = \sin(0) + C = C.$$

Since $x(0) = 2$, we obtain $C = 2$. Therefore, the unique solution satisfying both the ODE and the initial condition is

$$x(t) = \sin(t) + 2.$$

In general, solving an ODE can be much more complicated than finding an antiderivative. Nevertheless, integration provides a fundamental connection between a differential equation and its solution.

More generally, we can specify a condition of the form

$$x(t_0) = x_0,$$

where t_0 is a given initial point and x_0 is the corresponding value of the unknown function. The is called an **initial condition**. This leads naturally to the concept of an **initial value problem (IVP)**, in which a differential equation is accompanied by a prescribed value of the unknown function at an initial point.

2 Initial Value Problem

The initial value problem for a first-order ordinary differential equation

$$\begin{cases} y' = f(t, y(t)), & t \in [a, b] \\ y(a) = y_a \end{cases}$$

For example, **the logistic equation** $y' = ay(1 - y)$.
In real model, y usually represents the population.

$$\text{I.V.P} \quad \begin{cases} y' = ay(1 - y), & t \in [0, T] \\ y(0) = y_0 \end{cases} \qquad \longrightarrow \qquad y(t) = 1 - \frac{1}{1 + \frac{y_0}{1-y_0} e^{at}}, \text{ if } y_0 \neq 1.$$

A wide majority of interesting D.E. have no closed-form solution. A much more common scenario is a D.E. with no explicit solution formula. For example,

$$\begin{cases} y' = 3e^{\frac{t^2}{2}} - t^2 - 2 \\ y(0) = 1 \end{cases}$$

When an ordinary differential equation (ODE) has no closed-form solution, we cannot express its solution using standard analytical formulas. In such cases, we must turn to **numerical methods** to approximate the solution at discrete points.

3 Euler's Method

Euler's method uses the derivative at the current point to estimate the solution at the next point. Recall the definition of a derivative

$$y'(t) = \lim_{h \to 0} \frac{y(t + h) - y(t)}{h}.$$

For a sufficiently small step size h, we approximate the derivative by a forward difference

$$y'(t) \approx \frac{y(t + h) - y(t)}{h}.$$

Rearranging gives

$$y(t + h) \approx y(t) + hy'(t).$$

Since $y'(t) = f(t, y(t))$, we obtain Euler's formula

$$y_{n+1} = y_n + hf(t_n, y_n),$$

3.1 Lipschitz constant

Definition: A function $f(t, y)$ satisfies the Lipschitz condition in the variable y on the rectangles $S = [a, b] \times [\alpha, \beta]$ if there exists a constant L (called the **Lipschitz constant**) such that

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2| \quad \text{for all } (t, y_1), (t, y_2) \text{ in } S.$$

Note: A useful condition is that if the partial derivative $\frac{\partial f}{\partial y}$ exists and is bounded on a suitable region, then f is Lipschitz continuous in y there. According to Mean value theorem

$$\frac{f(t, y_1) - f(t, y_2)}{y_1 - y_2} = \frac{\partial f}{\partial y}(t, c)$$

where c is between y_1 and y_2. Then L can be taken as the maximum of $|\frac{\partial f}{\partial y}(t, c)|$ on S.

Example 1: A function that satisfies the condition. Consider

$$f(t, y) = 2y.$$

Then

$$|f(t, y_1) - f(t, y_2)| = 2|y_1 - y_2|.$$

Therefore, f is **globally** Lipschitz continuous in y, with $L = 2$.

Example 2: A function that is **locally** Lipschitz. Consider

$$f(t, y) = y^2.$$

We have

$$|y_1^2 - y_2^2| = |y_1 - y_2||y_1 + y_2|.$$

If $|y_1|$, $|y_2| \leq M$, then

$$|y_1^2 - y_2^2| \leq 2M|y_1 - y_2|.$$

Thus, f is Lipschitz continuous on every bounded interval in y, and **locally** Lipschitz on the entire real line.

Example 3: A continuous function that is not locally Lipschitz. Consider

$$f(t, y) = \sqrt{|y|}.$$

At $y = 0$, compare $y_1 = 0$ and $y_2 = y > 0$. The Lipschitz condition would require

$$\sqrt{y} \leq Ly,$$

or equivalently,

$$\frac{1}{\sqrt{y}} \leq L.$$

As y approaches zero, the left-hand side becomes unbounded. No finite L can satisfy the condition in a neighborhood of zero. So f is continuous but not locally Lipschitz at $y = 0$.

3.2 Existence and Uniqueness Theorem (E.U.T)

Definition: Assume that $f(t,y)$ is Lipschitz in the variable y on $[a,b]\times[\alpha,\beta]$ and $y_a \in (\alpha,\beta)$. Then $\exists c \in (a,b)$ such that I.V.P. $\begin{cases} y' = f(t,y) \\ y(a) = y_a \end{cases}$ has exactly one solution $y(t)$ for $t \in [a,c]$. Moreover, if f is Lipschitz on $[a,b]\times(-\infty,\infty)$, then there exists a unique solution on $[a,b]$.

Example 1: Consider $\begin{cases} y' = y^2 \\ y(0) = 1 \end{cases}$.

Since $f(t,y) = y^2$ we know $\frac{\partial f}{\partial y} = 2y$. $L = \max|2y| = 20$ on $[0,2]\times[-10,10]$. E.U.T. guarantees $y(t)$ uniquely exists on $[0,c]$, $c>0$ but not guarantees on the entire $[0,2]$.

In fact, $y(t) = \frac{1}{1-t}$. So, the solution exists on $[0,c]$ for any $0<c<1$.

Note: Since$\frac{dy}{y^2} = dt$, we obtain $-\frac{1}{y} = t + C$. Substituting the initial condition, so $C = -1$. Thus, $-\frac{1}{y} = t-1$ which can be rewritten $y = \frac{1}{1-t}$.

Example 2: Consider $\begin{cases} y' = y^2 \\ y(0) = y_0 \end{cases}$.

From previous example we know that $f(t,y) = y^2$ is Lipschitz continuous on $[-a,a]$ for all $a>0$.

Case 1: $y_0 \equiv 0$. Then $y(t) = 0$ is a solution.

Case 2: $y_0 \neq 0$. Then $y(t) = \frac{1}{y_0^{-1} - t}$.

$$\begin{cases} \text{If } y_0 > 0, \text{ then } y(t) = \frac{1}{y_0^{-1}-t} \text{ for } t \in (-\infty, \frac{1}{y_0}) \\ \text{If } y_0 < 0, \text{ then } y(t) = \frac{1}{y_0^{-1}-t} \text{ for } t \in (\frac{1}{y_0}, \infty) \end{cases}$$

Example 3: Consider $\begin{cases} y' = \begin{cases} \sqrt{y}, y \geq 0 \\ 0, y \leq 0 \end{cases} \\ y(0) = 0 \end{cases}$.

Case 1: $y(t) = \begin{cases} \frac{(t-c)^2}{4}, t \geq c \\ 0, t \leq c \end{cases}$ is a solution for any $c \geq 0$.

Case 2: $y(t) \equiv 0$ is also a solution.

Thus, in this example, there are infinitely many solutions.

Note: $\frac{dy}{\sqrt{y}} = dt$. We can write $\int y^{-1/2} dy = \int 1 dt$. So, $2\sqrt{y} = t + C$. We have $y = \frac{(t+C)^2}{4}$. Set $C = -c$, we have $y = \frac{(t-c)^2}{4}$

3.3 Continuous dependence of solutions on initial conditions

Definition: Assume that $f(t,y)$ is Lipschitz in variable y on $S=[a,b]\times[\alpha,\beta]$. If $y(t)$ and $z(t)$ are solutions of $\begin{cases} y'=f(t,y) \\ y(a)=y_a \end{cases}$ and $\begin{cases} z'=f(t,z) \\ z(a)=z_a \end{cases}$, respectively, then

$$|y(t)-z(t)| \le e^{L(t-a)}|y(a)-z(a)|.$$

Proof: By the uniqueness of solution, $y(a)=z(a)$ implies $y(t)=z(t)$. We may assume $y(a)\neq z(a)$. In this case, the uniqueness implies $y(t)\neq z(t), \forall t$.
Define $u(t)=y(t)-z(t)$, then $u(t)>0$ or $u(t)<0$. Because $|y(t)-z(t)|$ only depends on $|u|$, we may assume $u>0$.

$$u'(t)=y'(t)-z'(t)=f(t,y(t))-f(t,z(t))$$

The Lipschitz condition implies

$$u' \le |u'| = |f(t,y(t)-f(t,z(t))| \le L|y(t)-z(t)| = L|u(t)| = Lu(t),$$

which means

$$(\ln u)' = \frac{u'}{u} \le L.$$

By the mean value theorem, $\frac{\ln u(t)-\ln u(a)}{t-a} \le L$ which simplifies to $\ln\frac{u(t)}{u(a)} \le L(t-a)$. Thus, $u(t)\le u(a)e^{L(t-a)}$.

3.4 Global error

Theorem: Assume $f(t,y)$ has a Lipschitz constant L for the variable y and y_i is the solution of IVP $\begin{cases} y'=f(t,y) \\ y(a)=y_a \end{cases}$ $a\le t\le b$ at t_i, w_i is the one-step ODE solver withe the local truncation error $e_i \le ch^{k+1}$. Then for $a\le t_i\le b$, the solver has global truncation error

$$g_i = |w_i-y_i| \le \frac{c}{L}h^k(e^{L(t_i-a)}-1).$$

Proof: Set $g_0=|w_0-y_0|=|y_a-y_a|=0$ and $g_1=e_1=|w_1-y_1|$.
Let $z(t)$ be the solution of $\begin{cases} y'=f(t,y) \\ y(t_1)=w_1 \end{cases}$, $t\in[t_1,t_2]$ and then $e_2=|w_2-z(t_2)|$.
By initial value continuous theorem, we have $|z(t_2)-y_2|\le e^{Lh}|w_1-y_1|$. Thus

$$\begin{aligned} g_2 &= |w_2-y_2| \\ &= |w_2-z(t_2)+z(t_2)-y_2| \\ &\le |w_2-z(t_2)|+|z(t_2)-y_2| \\ &\le e_2+e^{Lh}g_1 \\ &= e_2+e^{Lh}e_1. \end{aligned}$$

Similarly,

$$\begin{aligned} g_3 &= |w_3 - y_3| \le e_3 + e^{Lh} g_2 \le e_3 + e^{Lh} e_2 + e^{2Lh} e_1. \\ &\vdots \\ g_i &= |w_i - y_i| \le e_i + e^{Lh} e_{i-1} + e^{2Lh} e_{i-2} + \cdots + e^{(n-1)Lh} e_1. \end{aligned}$$

Assume the local truncation error $e_i \le c \cdot h^{k+1}$. Then

$$\begin{aligned} g_i &\le ch^{k+1}[1 + e^{Lh} + \cdots + e^{(i+1)Lh}] \\ &= ch^{k+1} \frac{e^{iLh-1}}{e^{Lh} - 1} \\ &\le ch^{k+1} \frac{e^{L(t_i - a)-1}}{Lh} \\ &= \frac{c}{L} h^k (e^{L(t_i-a)-1}). \end{aligned}$$

Note:
$i \cdot h = (t_i - a)$
$1 + x \le e^x$ which means $1 + (Lh) \le e^{Lh}$. Then $Lh \le e^{Lh} - 1$.

3.5 Euler's Method Error

Assume $y \in \mathbb{C}^2$ the exact solution at $t_{i+1} = t_i + h$ is

$$y(t_i + h) = y(t_i) + h \cdot y'(t_i) + \frac{h^2}{2} y''(c), t_i \le c \le t_{i+1}.$$

Assume $y(t_i) = w_i$ and $y'(t_i) = f(t_i, w_i)$. Then

$$y(t_{i+1}) = w_i + h \cdot f(t_i, w_i) + \frac{h^2}{2} y''(c).$$

Meanwhile, Euler's method says $w_{i+1} = w_i + h \cdot f(t_i, w_i)$ which means

$$e_{i+1} = |w_{i+1} - y(t_{i+1})| = \frac{h^2}{2} |y''(c)| \le \frac{M}{2} h^2, \text{ where } M \equiv \max_{[t_i, t_{i+1}]} y''(x).$$

Corollary: Assume $f(t, y)$ has a Lipshitz constant L in variably y. The global truncation error from Euler's method satisfies

$$|w_i - y_i| \le \frac{Mh}{2L} (e^{L(t_i - a)-1}), M = \max_{t \in [a,b]} y''(t).$$

4 The Explicit Trapezoid Method (Heun Method)

Consider an initial value problem

$$\begin{cases} y' = f(t, y(t)) \\ y(t_0) = y_0 \end{cases}.$$

Euler's method approximates the solution using the slope at the beginning of each time step,

$$y_{n+1} = y_n + hf(t_n, y_n).$$

The weakness is that it uses only one slope. The Explicit Trapezoidal Method improves this by using both the slope at the beginning and an estimated slope at the end of the step.

The key idea is to approximate the integral of the derivative over one time step using the area of a trapezoid. Since $y' = f(t, y)$, integrating from t_n to t_{n+1} gives,

$$y(t_{n+1}) = y_n + \int_{t_n}^{t_{n+1}} f(t, y(t))dt.$$

We approximate this integral using the average of the two endpoint slopes multiplied by the step size.

4.1 Deriving the formula

For an initial value problem $y' = f(t, y)$ with $y(t_0) = y_0$, each time step h proceeds as follows:

Predictor Step (Euler): Calculate an intermediate approximation w^*_{i+1}.

$$w^*_{i+1} = w_i + hf(t, w_i).$$

Corrector Step (Trapezoidal Average): Use the average of the slope at the starting point and the slope evaluated at the predicted endpoint to obtain the corrected value.

$$w_{i+1} = w_i + \frac{h}{2}[f(t_i, w_i) + f(t_{i+1}, w_{i+1})].$$

Thus, we can write

$$\begin{cases} w_0 = y_0 \\ w_{i+1} = w_i + \frac{h}{2}[f(t_i, w_i) + f(t_i + h, w_i + h \cdot f(t_i, w_i))] \end{cases}$$

Example: Consider the initial value problem

$$y' = y + t, \qquad y(0) = 1$$

with step size $h = 0.1$. The exact solution is $y = 2e^t - t - 1$

Step1 - Predictor (Euler step) : $1 + (0.1) * (1 + 0) = 1.1$.

Step2 - Trapezoidal correction : $1 + (0.1/2) * (1 + (1.1 + 0.1)) = 1.11$.

The numerical approximation is 1.11 and exact value is $(2e^{0.1} - 0.1 - 1) \approx 1.110341836$.
So absolute error ≈ 0.000341836.

In the special case $y' = f(t)$, using the trapezoid rule to approximate $\int_{t_i}^{t_i+h} f(t)dt$ is

$$\frac{h}{2}[f(t_i) + f(t_i + h)].$$

Thus, the trapezoid method in the case is $w_{i+1} = w_i + \frac{h}{2}[f(t_i) + f(t_i + h)]$.
We do not need an Euler predictor because the derivative depends only on time and can be evaluated directly at both endpoints of the time interval.

For example: Consider the initial value problem $y' = 3t^2, y(0) = 1$, with step size $h = 0.1$.

Using the formula directly

$$w_1 = 1 + \frac{0.1}{2}[0 + 3 \cdot 0.1^2] = 1.0015.$$

The exact solution is $y(t) = t^3 + 1$, so the exact value at $t = 0.1$ is 1.001. Thus, the error is 0.0005. This numerical result is close to the exact solution.

4.2 Trapezoid Method Error

Assume $y \in \mathbf{C}^3$ and $y' = f(t, y)$ then

$$y'' = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial y}y' = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial y}f(t, y).$$

We have $y_{i+1} = y_i + hy'(t_i) + \frac{h^2}{2}y''(t_i) + \frac{h^3}{6}y'''(\xi_i)$, where $t_i < \xi_i < t_{i+1}$. Then, we can write

$$y_{i+1} = y_i + hy'(t_i) + \frac{h^2}{2}\left[\frac{\partial f}{\partial t}(t_i, y_i) + \frac{\partial f}{\partial y}(t_i, y_i)f(t_i, y_i)\right] + \frac{h^3}{6}y'''(\xi_i) \qquad -(*)$$

Using 2-dimensional Taylor expansion,

$$f(t_i + h, y_i + hf(t_i, y_i)) = f(t_i, y_i) + h\frac{\partial f}{\partial t}(t_i, y_i) + hf(t_i, y_i)\frac{\partial f}{\partial y}(t_i, y_i) + O(h^2).$$

Thus the trapezoid method can be written

$$\begin{aligned} w_{i+1} &= y_i + \frac{h}{2}[f(t_i, y_i) + f(t_i + h, y_i + hf(t_i, y_i))] \\ &= y_i + \frac{h}{2}f(t_i, y_i) + \frac{h}{2}[f(t_i, y_i) + h\frac{\partial f}{\partial t}(t_i, y_i) + hf(t_i, y_i)\frac{\partial f}{\partial y}(t_i, y_i) + O(h^2)] \\ &= y_i + hf(t_i, y_i) + \frac{h^2}{2}[\frac{\partial f}{\partial t}(t_i, y_i) + f(t_i, y_i)\frac{\partial f}{\partial y}(t_i, y_i)] + O(h^3) \qquad -(**) \end{aligned}$$

Let (*)-(**), which means $y_{i+1} - w_{i+1} = O(h^3)$.

Therefore, its one-step local truncation error is of order 3. Furthermore, by the theorem we mentioned global error in section 3.4, the global error of trapezoid method is of order 2.

4.3 Midpoint Method

Evaluates the slope at the estimated midpoint and uses it for the entire step.

$$\begin{cases} w_0 = y_0 \\ w_{i+1} = w_i + hf(t_i + \frac{h}{2}, w_i + \frac{h}{2}f(t_i, w_i)) \end{cases}$$

Comparison with the Explicit Trapezoidal Method, which estimates the slope at the predicted endpoint, then averages it with the initial slope.

$$\begin{cases} w_0 = y_0 \\ w_{i+1} = w_i + \frac{h}{2}f(t_i, w_i) + \frac{h}{2}[f(t_i + h, w_i + hf(t_i, w_i))] \end{cases}$$

4.4 Midpoint Method Error

$$\begin{aligned} y_{i+1} &= y_i + hf(t_i, y_i) + \frac{h^2}{2}\left[\frac{\partial f}{\partial t}(t_i, y_i) + \frac{\partial f}{\partial y}(t_i, y_i)f(t_i, y_i)\right] + \frac{h^3}{6}y'''(\xi_i) - (*) \\ w_i &= y_i + h\left[f(t_i + \frac{h}{2}, y_i + \frac{h}{2}f(t_i, y_i))\right] \\ &= y_i + h\left[f(t_i, y_i) + \frac{h}{2}\frac{\partial f}{\partial t}(t_i, y_i) + \frac{h}{2}f(t_i, y_i)\frac{\partial f}{\partial y}(t_i, y_i) + O(h^2)\right] - (**) \end{aligned}$$

Let (*)-(**), we have $y_{i+1} - w_{i+1} = O(h^3)$.

Therefore, its one-step local truncation error is of order 3. Furthermore, by the theorem we mentioned global error in section 3.4, the global error of midpoint method is of order 2. We got the same conclusion as Trapezoid method.

5 Runge-Kutta Method

Define: Each evaluation of the function $f(t, x)$ in the differential equation $\dot{x} = f(t, x)$ is called a **stage** of the method.

The explicit trapezoidal method and the midpoint method are both two-stage, second-order Runge–Kutta methods, having form

$$w_{i+1} = w_i + h(1 - \frac{1}{2\alpha})f(t_i, w_i) + \frac{h}{2\alpha}f(t_i + \alpha h, w_i + \alpha hf(t_i, w_i)), \alpha \neq 0$$

When $\alpha = 1$ is explicit trapezoid method, when $\alpha = \frac{1}{2}$ is midpoint method.

Classical fourth-order Runge–Kutta method can be written as

$$\begin{aligned} w_{i+1} &= w_i + \frac{h}{6}(s_1 + 2s_2 + 2s_3 + s_4) \\ s_1 &= f(t_i, w_i) \\ s_2 &= f(t_i + \frac{h}{2}, w_i + \frac{h}{2}s_1) \\ s_3 &= f(t_i + \frac{h}{2}, w_i + \frac{h}{2}s_2) \\ s_4 &= f(t_i + h, w_i + hs_3), \end{aligned}$$

where s_1 is the slope at t_i, s_2 is the slope used in the midpoint method, s_3 is the improved slope at midpoint, and s_4 is the improved slope at t_{i+1}.

5.1 s-stage Runge-Kutta Methods

$$\begin{cases} w_0 = y_a \\ w_{i+1} = w_i + h\phi(t_i, w_i) \end{cases}$$

where

$$\begin{aligned} \phi(t_i, w_i) &= \alpha_1 k_1 + \alpha_2 k_2 + \cdots + \alpha_s k_s \\ k_1 &= f(t_i, w_i) \\ k_2 &= f(t_i + \beta_2 h, w_i + h\gamma_{2,1} k_1) \\ k_3 &= f(t_i + \beta_3 h, w_i + h(\gamma_{3,1} k_1 + \gamma_{3,2} k_2)) \\ k_4 &= f(t_i + \beta_4 h, w_i + h(\gamma_{4,1} k_1 + \gamma_{4,2} k_2 + \gamma_{4,3} k_3)) \\ &\vdots \\ k_s &= f(t_i + \beta_s h, w_i + h(\gamma_{s,1} k_1 + \gamma_{s,2} k_2 + \gamma_{s,3} k_3) + \cdots + \gamma_{s,s-1} k_{s-1}). \end{aligned}$$

Note:
C1: For being consistent, the condition $\alpha_1 + \alpha_2 + \cdots + \alpha_s = 1$ is imposed.
C2: The condition $\gamma_{j,1} + \gamma_{j,2} + \cdots + \gamma_{j,j-1} = \beta_j$ for $j = 2, 3, \ldots, s$
is equivalent to $k_j(t_i, w_i, h) = z'(t_i + \beta_j h) + O(h^2)$, for $j = 2, 3, \ldots, s$.

2-stage RK:

$$\begin{cases} w_0 = y_a \\ w_{i+1} = w_i + h(\alpha_1 k_1 + \alpha_2 k_2) \end{cases} \quad \text{,where} \begin{cases} k_1 = f(t_i, w_i) \\ k_2 = f(t_i + \beta_2 h, w_i + \gamma_{2,1} h k_1) \end{cases}$$

and $\alpha_1 + \alpha_2 = 1$, $\beta_2 = \gamma_{2,1}$.
We can write

$$\begin{cases} w_0 = y_a \\ w_{i+1} = w_i + h\phi(t_i, w_i) \end{cases} \quad \text{,where } \phi(t_i, w_i) = (1-\alpha) f(t_i, w_i) + \alpha f(t_i + \beta h, w_i + \beta h f(t_i, w_i)).$$

Midpoint Method: $\alpha = 1, \beta = \frac{1}{2}$

$$\begin{cases} w_0 = y_0 \\ w_{i+1} = w_i + h f(t_i + \frac{h}{2}, w_i + \frac{h}{2} f(t_i, w_i)) \end{cases}$$

Trapezoid Method: $\alpha = \frac{1}{2}, \beta = 1$

$$\begin{cases} w_0 = y_0 \\ w_{i+1} = w_i + \frac{h}{2} \left[f(t_i, w_i) + f(t_i + h, w_i + h f(t_i, w_i)) \right] \end{cases}$$

(2nd order Heun's) Method: $\alpha = \frac{3}{4}, \beta = \frac{2}{3}$

$$\begin{cases} w_0 = y_0 \\ w_{i+1} = w_i + \frac{h}{4} \left[f(t_i, w_i) + 3 f(t_i + \frac{2}{3} h, w_i + \frac{2}{3} h f(t_i, w_i)) \right] \end{cases}$$

3-stage RK:

$$\begin{cases} w_0 = y_a \\ w_{i+1} = w_i + h\phi(t_i, w_i) \end{cases},$$

where

$$\begin{aligned} \phi(t_i, w_i) &= \alpha_1 k_1 + \alpha_2 k_2 + \alpha_3 k_3 \\ k_1 &= f(t_i, w_i) \\ k_2 &= f(t_i + \beta_2 h, w_i + \gamma_{2,1} h k_1) \\ k_3 &= f(t_i + \beta_3 h, w_i + \gamma_{3,1} h k_1 + \gamma_{3,2} h k_2) \end{aligned}$$

and $\alpha_1 + \alpha_2 + \alpha_3 = 1$, $\begin{cases} \beta_2 = \gamma_{2,1} \\ \beta_3 = \gamma_{3.1} + \gamma_{3,2} \end{cases}$.

We can write

$$\begin{cases} w_0 = y_a \\ w_{i+1} = w_i + h\phi(t_i, w_i) \end{cases} \text{,where}$$

$$\begin{aligned} \phi(t_i, w_i) = (1 - \alpha_2 - \alpha_3) f(t_i, w_i) + \alpha_2 f(t_i + \beta h, w_i + \beta h f(t_i, w_i)) \\ + \alpha_3 f(t_i, (\gamma_1 + \gamma_2)h, w_i + \gamma_1 h f(t_i, w_i) + \gamma_2 h f(t_i + \beta h, w_i + \beta h f(t_i, w_i))). \end{aligned}$$

(3nd order Heun's) Method: $\alpha_2 = 0, \alpha_3 = \frac{3}{4}, \beta = \frac{1}{3}, \gamma_1 = 0, \gamma_2 = \frac{2}{3}$

$$\begin{cases} w_0 = y_0 \\ w_{i+1} = w_i + \frac{h}{4}\left[f(t_i, w_i) + 3f(t_i + \frac{2}{3}h, w_i + \frac{2}{3}hf(t_i + \frac{h}{3}, w_i + \frac{h}{3}f(t_i, w_i)))\right] \end{cases}$$

5.2 Runge-Kutta Method Error

In RK4, we have

$$\begin{aligned} y_{i+1} = y_i + hf + \frac{h^2}{2}\left[f_t + f_y f\right] + \frac{h^3}{6}\left[f_{tt} + 2ff_{ty} + f^2 f_{yy} + f_t f_y + f f_y^2\right] \\ + \frac{h^4}{24}\left[f_{ttt} + 3ff_{tty} + 3f^2 f_{tyy} + f^3 f_{yyy} + f_y f_{tt} + 3f_t f_{ty} + 3ff_t f_{yy} + 4ff_y f_{ty} + 6f^2 f_y f_{yy} + f_t f_y^2 + f f_y^3\right] \\ + \frac{h^5}{120} y^{(5)}(\xi_i) - (*) \end{aligned}$$

The numerical solution is

$$\begin{aligned} w_{i+1} &= w_i + \frac{h}{6}(s_1 + 2s_2 + 2s_3 + s_4) \\ s_1 &= f(t_i, w_i) \\ s_2 &= f(t_i + \frac{h}{2}, w_i + \frac{h}{2}s_1) \\ s_3 &= f(t_i + \frac{h}{2}, w_i + \frac{h}{2}s_2) \\ s_4 &= f(t_i + h, w_i + hs_3), \end{aligned}$$

Using Taylor's expansion

$$\begin{aligned} f(t+\Delta t, y+\Delta y) &= f + f_t\Delta t + f_y\Delta y \\ &\quad + \frac{1}{2}(f_{tt}\Delta t^2 + 2f_{ty}\Delta t\Delta y + f_{yy}\Delta y^2) \\ &\quad + \frac{1}{6}(f_{ttt}\Delta t^3 + 3f_{tty}\Delta t^2\Delta y + 3f_{tyy}\Delta t\Delta y^2 + f_{yyy}\Delta y^3) + O(h^4). \end{aligned}$$

s2: We know $\Delta t = \frac{h}{2}$ and $\Delta y = \frac{h}{2}f$. Thus,

$$\begin{aligned} s_2 &= f + \frac{h}{2}(f_t + ff_y) \\ &\quad + \frac{h^2}{8}(f_{tt} + 2ff_{ty} + f^2 f_{yy}) \\ &\quad + \frac{h^3}{48}(f_{ttt} + 3ff_{tty} + 3f^2 f_{tyy} + f^3 f_{yyy}) + O(h^4). \end{aligned}$$

Let

$$\begin{aligned} A &= f_t + ff_y \\ B &= f_{tt} + 2ff_{ty} + f^2 f_{yy} \\ C &= f_{ttt} + 3ff_{tty} + 3f^2 f_{tyy} + f^3 f_{yyy}. \end{aligned}$$

We can rewrite

$$s_2 = f + \frac{h}{2}A + \frac{h^2}{8}B + \frac{h^3}{48}C + O(h^4).$$

s3: We know $\Delta t = \frac{h}{2}$ and $\Delta y = \frac{h}{2}s_2 = \frac{h}{2}f + \frac{h^2}{4}A + \frac{h^3}{16}B + \frac{h^4}{96}C + O(h^5)$. Thus,

$$s_3 = f + \frac{h}{2}A + \frac{h^2}{8}B + \frac{h^2}{4}f_yA + \frac{h^3}{48}C + \frac{h^3}{16}f_yB + \frac{h^3}{8}f_{ty}A + \frac{h^3}{8}ff_{yy}A + O(h^4).$$

s4: We know $\Delta t = h$ and $\Delta y = hs_3 = hf + \frac{h^2}{2}A + h^3\left(\frac{1}{8}B + \frac{1}{4}f_yA\right) + O(h^4)$. Thus,

$$s_4 = f + hA + \frac{h^2}{2}B + \frac{h^2}{2}f_yA + \frac{h^3}{6}C + \frac{h^3}{8}f_yB + \frac{h^3}{4}f_y^2A + \frac{h^3}{2}f_{ty}A + \frac{h^3}{2}ff_{yy}A + O(h^4).$$

So, we have

$$\begin{aligned} w_{i+1} &= y_i + hf + \frac{h^2}{2}\left[f_t + f_yf\right] + \frac{h^3}{6}\left[f_{tt} + 2ff_{ty} + f^2f_{yy} + f_tf_y + ff_y^2\right] \\ &\quad + \frac{h^4}{24}\left[f_{ttt} + 3ff_{tty} + 3f^2f_{tyy} + f^3f_{yyy} + f_yf_{tt} + 3f_tf_{ty} + 3ff_tf_{yy} + 5ff_yf_{ty} + 4f^2f_yf_{yy} + f_tf_y^2 + ff_y^3\right] \\ &\quad + O(h^5) - (**) \end{aligned}$$

Therefore, its one-step local truncation error is of order 5. Furthermore, by the theorem we mentioned global error in section 3.4, the global error of RK4 is of order 4.

Note:

$$(1)\qquad f_t\Delta t + f_y\Delta y = \frac{h}{2}f_t + f_y\left(\frac{h}{2}f + \frac{h^2}{4}A + \frac{h^3}{16}B + O(h^4)\right)$$
$$= \frac{h}{2}(f_t + ff_y) + \frac{h^2}{4}f_yA + \frac{h^3}{16}f_yB + O(h^4).$$

$$(2)\qquad \frac{1}{2}f_{tt}\Delta t^2 = \frac{h^2}{8}f_{tt}.$$

$$(3)\qquad f_{ty}\Delta t\Delta y = f_{ty}\frac{h}{2}\left(\frac{h}{2}f + \frac{h^2}{4}A + \frac{h^3}{16}B + O(h^4)\right)$$
$$= \frac{h^2}{4}ff_{ty} + \frac{h^3}{8}f_{ty}A + O(h^4).$$

$$(4)\qquad \frac{1}{2}f_{yy}\Delta y^2 = \frac{1}{2}f_{yy}\left(\frac{h}{2}f + \frac{h^2}{4}A + \frac{h^3}{16}B + O(h^4)\right)^2$$
$$= \frac{1}{2}f_{yy}\left(\left(\frac{h}{2}f\right)^2 + 2\left(\frac{h}{2}f\right)\left(\frac{h^2}{4}A\right) + \left(\frac{h^2}{4}A\right)^2 + 2\left(\frac{h}{2}f\right)\left(\frac{h^3}{16}B\right) + O(h^5)\right)$$
$$= \frac{h^2}{8}f^2f_{yy} + \frac{h^3}{8}ff_{yy}A + O(h^4).$$

$$(5)\qquad \frac{1}{6}f_{ttt}\Delta t^3 = \frac{h^3}{48}f_{ttt}.$$

$$(6)\qquad \frac{1}{2}f_{tty}\Delta t^2\Delta y = \frac{1}{2}f_{tty}\left(\frac{h^2}{4}\left(\frac{h}{2}f + O(h^2)\right)\right)$$
$$= \frac{h^3}{16}ff_{tty} + O(h^4).$$

$$(7)\qquad \frac{1}{2}f_{tyy}\Delta t\Delta y^2 = \frac{1}{2}f_{tyy}\left(\frac{h}{2}\left(\frac{h^2}{4}f^2 + O(h^3)\right)\right)$$
$$= \frac{h^3}{16}f^2f_{tyy} + O(h^4).$$

$$(8)\qquad \frac{1}{6}f_{yyy}\Delta y^3 = \frac{1}{6}f_{yyy}\left(\frac{h^3}{8}f^3 + O(h^4)\right)$$
$$= \frac{h^3}{48}f^3f_{yyy} + O(h^4).$$

Add from (1) to (8), we have

$$s_3 = f + \frac{h}{2}A + \frac{h^2}{8}B + \frac{h^2}{4}f_yA + \frac{h^3}{48}C + \frac{h^3}{16}f_yB + \frac{h^3}{8}f_{ty}A + \frac{h^3}{8}ff_{yy}A + O(h^4).$$

We can implement the same way to get s_4.

6 Boundary Value Problem

In an initial value problem (IVP), we know the solution and its derivatives at the initial point. In a boundary value problem (BVP), conditions are imposed at **the boundaries of the interval**.

The key difference is that a BVP must satisfy conditions at both ends of the interval, so we cannot generally solve it by simply starting at the left endpoint and marching forward as we do with an IVP. Consider

$$\begin{cases} -u'' + u = f(x), & x \in [0,1] \\ u(0) = u(1) = 0. \end{cases}$$

Let $V = \{v(x) \in c[0,1] | v' \text{ is piecewise and bounded in } [0,1] \text{ with } v(0) = v(1) = 0\}$. We multiply both sides by a test function $v(x)$.

$$[-u''(x) + u(x)]v(x) = f(x)v(x).$$

Integrate from 0 to 1

$$\int_0^1 [-u''(x) + u(x)]v(x)dx = \int_0^1 f(x)v(x)dx.$$

Expanding the left-hand side gives

$$\int_0^1 -u''(x)v(x) + u(x)v(x)dx = \int_0^1 f(x)v(x)dx. - (*)$$

By the integration by parts, we have

$$\int_0^1 -u''(x)v(x)dx = \int_0^1 -v(x)du'(x) = [-v(x)u'(x)]|_0^1 - \int_0^1 -u'(x)v'(x)dx = 0 + \int_0^1 u'(x)v'(x)dx.$$

Substitute into $(*)$, we have

$$\int_0^1 [u'(x)v'(x) + u(x)v(x)]dx = \int_0^1 f(x)v(x)dx. - (**)$$

Let $B(u,v) \equiv \int_0^1 u'(x)v'(x) + u(x)v(x)dx$ and $\langle f, v\rangle \equiv \int_0^1 f(x)v(x)dx$.

Then $(**)$ become $B(u,v) = \langle f, v\rangle$ which is called the **weak formulation** of BVP.

6.1 Finite Difference Method (FDM)

The finite difference method approximates derivatives using function values at nearby grid points. By using different combinations of these values, we can approximate derivatives of different orders. For a n-times differentiable function, by Taylor's theorem the Taylor series expansion is given as

$$u(x_0+h) = u(x_0) + \frac{u^{(1)}(x_0)}{1!}h + \frac{u^{(2)}(x_0)}{2!}h^2 + \frac{u^{(3)}(x_0)}{3!}h^3 + \cdots + \frac{u^{(n)}(x_0)}{n!}h^n + R_n(x).$$

First order

$$u(x_i+h) = u(x_i) + u'(x_i)h + O(h^2).$$

we obtain the forward difference approximation

$$u'(x_i) \approx \frac{u(x_i+h) - u(x_i)}{h}.$$

We can write

$$u'(x_i) \approx \frac{u_{i+1} - u_i}{h}.$$

A more accurate central difference formula is

$$u'(x_i) \approx \frac{u_{i+1} - u_{i-1}}{2h}.$$

which has a truncation error $O(h^2)$.

Second order

$$u(x_i+h) = u_i + u_i'h + \frac{u_i''}{2}h^2 + \frac{u_i'''}{6}h^3 + O(h^4).$$
$$u(x_i-h) = u_i - u_i'h + \frac{u_i''}{2}h^2 - \frac{u_i'''}{6}h^3 + O(h^4).$$

Adding these equations and rearranging gives

$$u''(x_i) \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}.$$

which has a truncation error $O(h^2)$.

Third and higher-order derivatives
The same principle can be extended to higher-order derivatives. For example, a central difference approximation to the third derivative is

$$u'''(x_i) \approx \frac{u_{i+2} - 2u_{i+1} + 2u_{i-1} - u_{i-2}}{2h^3}$$

with truncation error $O(h^2)$.

Discretizations: Consider the following boundary value problem:

$$\begin{cases} -u''(x) + u(x) = f(x), & 0 < x < 1, \\ u(0) = u(1) = 0. \end{cases}$$

Divide the interval $[0, 1]$ into N equally spaced sub-intervals with step size

$$h = \frac{1}{N}.$$

The grid points are defined as

$$x_i = ih, \qquad i = 0, 1, \ldots, N.$$

Let u_i denote the numerical approximation to $u(x_i)$. The boundary conditions give

$$u_0 = 0, \qquad u_N = 0.$$

At each interior grid point x_i, where $i = 1, \ldots, N-1$, we approximate the second derivative using the central difference formula

$$u''(x_i) \approx \frac{u_{i-1} - 2u_i + u_{i+1}}{h^2}.$$

Substituting this approximation into the differential equation yields

$$-\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} + u_i = f(x_i).$$

Rearranging, we obtain

$$-\frac{1}{h^2}u_{i-1} + \left(\frac{2}{h^2} + 1\right) u_i - \frac{1}{h^2}u_{i+1} = f(x_i).$$

This equation holds at every interior grid point.

Example
For example, let $N = 4$, so $h = \frac{1}{4}$. There are three interior unknowns u_1, u_2, u_3. The discrete equations become

$$\begin{aligned} 33u_1 - 16u_2 &= f(x_1) \\ -16u_1 + 33u_2 - 16u_3 &= f(x_2) \\ -16u_2 + 33u_3 &= f(x_3) \end{aligned}$$

Therefore,

$$\begin{bmatrix} 33 & -16 & 0 \\ -16 & 33 & -16 \\ 0 & -16 & 33 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} f(x_1) \\ f(x_2) \\ f(x_3) \end{bmatrix}$$

Notice that the matrix is **tridiagonal**: only the main diagonal and the two adjacent diagonals contain nonzero entries. We have transformed a differential equation into a linear algebra problem, $A\mathbf{x} = \mathbf{b}$. Since the resulting matrix is tridiagonal, we can efficiently solve the system using the **Thomas algorithm** (see Appendix A.2).

6.2 Finite Element Method (FEM)

Recall the weak form of BVP

$$B(u,v) = \langle f, v\rangle$$

The Finite Element Method (FEM) is to construct a finite-dimensional subspace V_n of V.
A simple case: Consider V_n consists of continuous, piecewise linear functions. Let

$$0 = x_0 < x_1 < x_2 < \cdots < x_n < x_{n+1} = 1.$$

Partition $[0,1]$ into subintervals $I_j = [x_{j-1}, x_j]$ of length $h_j = x_j - x_{j-1}$ for $j = 1, 2, \ldots, n+1$.
Define $\phi_j(x)$ to be continuous, piecewise linear function with

$$\phi_j(x_i) = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Then

$$\phi_j(x) = \begin{cases} \frac{x - x_{j-1}}{h_j}, & x_{j-1} \leq x \leq x_j \\ \frac{x_{j+1} - x}{h_{j+1}}, & x_j \leq x \leq x_{j+1} \\ 0, & \text{otherwise} \end{cases}$$

Let $\phi_1, \phi_2, \ldots, \phi_n$ be a basis of V_n and approximate $u(x)$ by $u_n(x) = \sum_{i=1}^n c_i \phi_i(x)$. Then, we have

$$B(u_n, \phi_j) = \langle f, \phi_j\rangle \quad \rightarrow \quad B(\sum_{i=1}^n c_i \phi_i, \phi_j) = \langle f, \phi_j\rangle$$

$B(\sum_{i=1}^n c_i \phi_i, \phi_j) = \langle f, \phi_j\rangle$ which can be written as

$$A\mathbf{x} = \mathbf{f}$$

where $A = (a_{jj})$ with $a_{ij} = B(\phi_i, \phi_j)$, $\mathbf{c} = (c_1, c_2, \ldots, c_n)^T$, and $\mathbf{f} = (f_j)$ with $f_j = \langle f, \phi_j\rangle$.

Recall $B(u,v) \equiv \int_0^1 u'(x)v'(x) + u(x)v(x)dx$, we have

$$a_{j,j} = \int_0^1 \phi_j'\phi_j' + \phi_j\phi_j dx = \int_{x_{j-1}}^{x_{j+1}} (\phi_j')^2 + (\phi_j)^2 dx$$

$$= \int_{x_{j-1}}^{x_j} \left[\left(\frac{x - x_{j-1}}{h_j}\right)'\right]^2 + \left(\frac{x - x_{j-1}}{h_j}\right)^2 dx + \int_{x_j}^{x_{j+1}} \left[\left(\frac{x_{j+1} - x}{h_{j+1}}\right)'\right]^2 + \left(\frac{x_{j+1} - x}{h_{j+1}}\right)^2 dx$$

$$= \int_{x_{j-1}}^{x_j} \left(\frac{1}{h_j}\right)^2 + \left(\frac{x - x_{j-1}}{h_j}\right)^2 dx + \int_{x_j}^{x_{j+1}} \left(\frac{-1}{h_{j+1}}\right)^2 + \left(\frac{x_{j+1} - x}{h_{j+1}}\right)^2 dx$$

$$= \left(\frac{1}{h_j}\right)^2 \left[x + \frac{1}{3}(x - x_{j-1})^3\right]_{x_{j-1}}^{x_j} + \left(\frac{1}{h_{j+1}}\right)^2 \left[x - \frac{1}{3}(x_{j+1} - x)^3\right]_{x_j}^{x_j+1}$$

$$= \left(\frac{1}{h_j} + \frac{h_j}{3}\right) + \left(\frac{1}{h_{j+1}} + \frac{h_{j+1}}{3}\right)$$

$$
\begin{aligned}
a_{j,j-1} &= \int_0^1 \phi_j'\phi_{j-1}' + \phi_j\phi_{j-1}dx = \int_{x_{j-1}}^{x_j} \phi_j'\phi_{j-1}' + \phi_j\phi_{j-1}dx \\
&= \int_{x_{j-1}}^{x_j} \left(\frac{x-x_{j-1}}{h_j}\right)'\left(\frac{x_j-x}{h_j}\right)' + \left(\frac{x-x_{j-1}}{h_j}\right)\left(\frac{x_j-x}{h_j}\right)dx \\
&= \int_{x_{j-1}}^{x_j} \frac{1}{h_j}\frac{-1}{h_j} + \left(\frac{(x-x_{j-1})(x_j-x)}{(h_j)^2}\right)dx \\
&= \left(\frac{1}{h_j}\right)^2 \int_{x_{j-1}}^{x_j} -1 + (x-x_{j-1})(x_j-x)dx \\
&= \frac{1}{(h_j)^2}\left(-h_j + \frac{1}{6}h_j^3\right) = -\frac{1}{h_j} + \frac{h_j}{6}
\end{aligned}
$$

Thus, $a_j = a_{j,j} = \left(\frac{1}{h_j} + \frac{h_j}{3}\right) + \left(\frac{1}{h_{j+1}} + \frac{h_{j+1}}{3}\right)$ and $b_j = a_{j,j-i} = a_{j-1,j} = -\frac{1}{h_j} + \frac{h_j}{6}$.

The equal space case : $h_j = h = \frac{1}{n+1}$ for all j. Then,

$$
a_j = \frac{2}{h} + \frac{2}{3}h, \quad b_j = -\frac{1}{h} + \frac{h}{6}
$$

Thus,

$$
A = \frac{1}{h}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & & \\ & & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}_{n*n} + \frac{h}{6}\begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & & \\ & & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{bmatrix}_{n*n}
$$

Recall $\int_0^1 [u'(x)v'(x) + u(x)v(x)]dx = \int_0^1 f(x)v(x)dx$.

$$
\begin{aligned}
b_j &= \int_0^1 f(x)\phi_j(x)dx \\
&= \int_{x_{j-1}}^{x_j} f(x)\phi_j(x)dx + \int_{x_j}^{x_{j+1}} f(x)\phi_j(x)dx
\end{aligned}
$$

If $f(x) = 1$, then

$$
b_j = \int_{x_{j-1}}^{x_j} \phi_j(x)dx + \int_{x_j}^{x_{j+1}} \phi_j(x)dx = \frac{h_j}{2} + \frac{h_{j+1}}{2} = h
$$

If $f(x) = x$, then

$$
b_j = \int_{x_{j-1}}^{x_j} x\phi_j(x)dx + \int_{x_j}^{x_{j+1}} x\phi_j(x)dx = hx_j
$$

A Appendix

A.1 Taylor's Theorem in two variables

$$f(t,y) = P_n(t,y) + R_n(t,y)$$

where

$$\begin{aligned} P_n(t,y) = f(t_0,y_0) &+ \left[(t-t_0)\frac{\partial f}{\partial t}(t_0,y_0) + (y-y_0)\frac{\partial f}{\partial y}(t_0,y_0)\right] \\ &+ \left[\frac{(t-t_0)^2}{2}\frac{\partial^2 f}{\partial t^2}(t_0,y_0) + (t-t_0)(y-y_0)\frac{\partial^2 f}{\partial t \partial y}(t_0,y_0) + \frac{(y-y_0)^2}{2}\frac{\partial^2 f}{\partial y^2}(t_0,y_0)\right] \\ &+ \cdots + \left[\frac{1}{n!}\sum_{j=0}^{n}\binom{n}{j}(t-t_0)^{n-j}(y-y_0)^j\frac{\partial^n f}{\partial t^{n-j}\partial y^j}(t_0,y_0)\right] \end{aligned}$$

and

$$R_n(t,y) = \frac{1}{(n+1)!}\sum_{j=0}^{n+1}\binom{n+1}{j}(t-t_0)^{n+1-j}(y-y_0)^j\frac{\partial^{n+1} f}{\partial t^{n+1-j}\partial y^j}(\xi,\mu).$$

$n = 2$:

$$f(t+\Delta t, y+\Delta y) = f(t,y) + f_t\Delta t + f_y\Delta y + \frac{1}{2}\left[f_{tt}(\Delta t)^2 + 2f_{ty}\Delta t\Delta y + f_{yy}(\Delta y)^2\right] + \ldots$$

$$R_2(t,y) = \frac{1}{6}\left[(\Delta t)^3 f_{ttt} + 3(\Delta t)^2\Delta y f_{tty} + 3\Delta t(\Delta y)^2 f_{tyy} + (\Delta y)^3 f_{yyy}\right](\xi,\mu).$$

$n = 3$:

$$\begin{aligned} f(t+\Delta t, y+\Delta y) = f(t,y) &+ f_t\Delta t + f_y\Delta y + \frac{1}{2}\left[f_{tt}(\Delta t)^2 + 2f_{ty}\Delta t\Delta y + f_{yy}(\Delta y)^2\right] \\ &+ \frac{1}{6}\left[(\Delta t)^3 f_{ttt} + 3(\Delta t)^2\Delta y f_{tty} + 3\Delta t(\Delta y)^2 f_{tyy} + (\Delta y)^3 f_{yyy}\right] + \ldots \end{aligned}$$

$$R_3(t,y) = \frac{1}{24}\left[(\Delta t)^4 f_{tttt} + 4(\Delta t)^3\Delta y f_{ttty} + 6(\Delta t)^2(\Delta y)^2 f_{ttyy} + 4(\Delta y)^3 f_{tyyy} + (\Delta y)^4 f_{yyyy}\right](\xi,\mu).$$

A.2 Thomas Algorithm

The Thomas algorithm is a direct method for solving tridiagonal linear systems. It is a specialized form of Gaussian elimination that exploits the structure of the coefficient matrix. Consider the linear system

$$A\mathbf{x} = \mathbf{d},$$

where A is an $n \times n$ tridiagonal matrix.

$$\begin{bmatrix} b_1 & c_1 & 0 & \cdots & 0 \\ a_2 & b_2 & c_2 & \ddots & \vdots \\ 0 & a_3 & b_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & c_{n-1} \\ 0 & \cdots & 0 & a_n & b_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{bmatrix}.$$

Equivalently, the system can be written as

$$a_i x_{i-1} + b_i x_i + c_i x_{i+1} = d_i, \qquad i = 1, \ldots, n,$$

where $a_1 = 0$ and $c_n = 0$.

For example

$$\left[\begin{array}{cccc|c} b_1 & c_1 & 0 & 0 & d_1 \\ a_2 & b_2 & c_2 & 0 & d_2 \\ 0 & a_3 & b_3 & c_3 & d_3 \\ 0 & 0 & a_4 & b_4 & d_4 \end{array}\right]$$

Forward Elimination
Initialize the modified coefficients:

$$\widetilde{c}_1 = \frac{c_1}{b_1}, \quad \widetilde{d}_1 = \frac{d_1}{b_1} \qquad \longrightarrow \qquad \left[\begin{array}{cccc|c} 1 & \widetilde{c}_1 & 0 & 0 & \widetilde{d}_1 \\ a_2 & b_2 & c_2 & 0 & d_2 \\ 0 & a_3 & b_3 & c_3 & d_3 \\ 0 & 0 & a_4 & b_4 & d_4 \end{array}\right]$$

$$m_2 = b_2 - a_2\widetilde{c}_1 \qquad \longrightarrow \qquad \left[\begin{array}{cccc|c} 1 & \widetilde{c}_1 & 0 & 0 & \widetilde{d}_1 \\ 0 & m_2 & c_2 & 0 & d_2 - a_2\widetilde{d}_1 \\ 0 & a_3 & b_3 & c_3 & d_3 \\ 0 & 0 & a_4 & b_4 & d_4 \end{array}\right]$$

$$\widetilde{c}_2 = \frac{c_2}{m_2}, \quad \widetilde{d}_2 = \frac{d_2 - a_2\widetilde{d}_1}{m_2} \qquad \longrightarrow \qquad \left[\begin{array}{cccc|c} 1 & \widetilde{c}_1 & 0 & 0 & \widetilde{d}_1 \\ 0 & 1 & \widetilde{c}_2 & 0 & \widetilde{d}_2 \\ 0 & a_3 & b_3 & c_3 & d_3 \\ 0 & 0 & a_4 & b_4 & d_4 \end{array}\right]$$

$$\vdots$$

$$\left[\begin{array}{cccc|c} 1 & \widetilde{c}_1 & 0 & 0 & \widetilde{d}_1 \\ 0 & 1 & \widetilde{c}_2 & 0 & \widetilde{d}_2 \\ 0 & 0 & 1 & \widetilde{c}_3 & \widetilde{d}_3 \\ 0 & 0 & 0 & 1 & \widetilde{d}_4 \end{array}\right]$$

Let $\widetilde{c}_1 = \frac{c_1}{b_1}, \widetilde{d}_1 = \frac{d_1}{b_1}$. For $i = 2, \ldots, n$, compute

$$m_i = b_i - a_i \widetilde{c}_{i-1},$$

and update

$$\widetilde{c}_i = \begin{cases} \dfrac{c_i}{m_i}, & i < n, \\ 0, & i = n, \end{cases}$$

together with

$$\widetilde{d}_i = \frac{d_i - a_i \widetilde{d}_{i-1}}{m_i}.$$

This process eliminates the lower diagonal entries while preserving the tridiagonal structure.

Backward Substitution
Starting with the last unknown, calculate

$$x_n = \widetilde{d}_n.$$

Then, for $i = n-1, n-2, \ldots, 1$, compute

$$x_i = \widetilde{d}_i - \widetilde{c}_i x_{i+1}.$$

The resulting vector $\mathbf{x}$ is the solution of the tridiagonal linear system.

Computational Complexity
The forward elimination and backward substitution each require $O(n)$ arithmetic operations. Therefore, the overall computational complexity is $O(n)$, and the storage requirement is $O(n)$ when the diagonal coefficients are stored in vectors.

The standard Thomas algorithm requires all elimination pivots m_i to be nonzero. Unlike Gaussian elimination with partial pivoting, it does not automatically handle zero or unsuitable pivots.